Chapter

Virtual Instrument for Single-Phase Control of Three-Phase Reactive Power Capacitive Compensators

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Abstract

This chapter proposes the use of LabVIEW in a very common application both in the field of power engineering technical education and in the industrial environment. It is about compensating the reactive power with the aim of increasing the power factor. Capacitive reactive power compensators for this purpose are provided with compensation steps. The connection and disconnection of these steps according to the load curve of the consumer and the power factor at which it operates are done by means of programmable controllers. Most capacitive compensators are designed to be connected balanced to the three-phase network, in delta connection. In the situation when the reactive capacitive compensators are designed for both power factor improvement and load balancing, single-phase connection of capacitor steps is needed. The chapter proposes a LabVIEW virtual instrument that, by means of National Instruments equipment, is designed for operating condition monitoring and single-phase command. The virtual instrument was tested in the laboratory on a model consisting of a RL consumer with step-adjustable inductive component and a step-adjustable capacitive compensator. The National Instruments hardware equipment used, the modeled consumer, and its related capacitive compensator are presented. Special attention, with details of realization, is given to the virtual instrument architecture.

Keywords: reactive power compensation, shunt capacitive compensator, data acquisition, virtual instrument, single-phase command

1. Introduction

In the field of power engineering, one of the methods of increasing the efficiency of transmission and distribution facilities, and implicitly of preserving energy resources, is the concern of improving the power factor. Having a high power factor in the energy system avoids the reactive power flow from power plants to consumers through lines, stations, and substations, reducing active energy losses to the minimum level of own technological consumption.

Power factor correction is generally done by shunt capacitive compensation made by stepped capacitor banks, which, together with their switching devices (contactors,

relays, static switching devices) and the automatic control, form the power reactive capacitive compensators [1]. The automatic control is done in most of the cases balanced on the three phases of the network by programmable logic controllers, the purpose of the compensation being the power factor improvement and the voltage regulation.

Another problem that leads to the reduction of the efficiency of the operation of electrical networks is their unbalanced regime. An unbalanced three-phase capacitive compensator can be used both for power factor compensation and for load balancing of unbalanced networks [2]. To be able to operate in this way, a single-phase command of the capacitive compensator is needed. A combined compensation strategy based on instantaneous reactive power theory for reactive power compensation and load balancing is also presented in Ref. [3]. Another category of devices, which, in addition to compensating the reactive power, also acts on the unbalances in the network, is the Static Var Compensator (SVC) type equipment [4–10]. They use power electronics components to control passive reactive circuit elements.

In addition to operating at a low power factor and under unbalanced regimes, the electrical networks also present other problems regarding the power quality such as harmonic regime, voltage dips, and flicker. A unitary solution for solving these problems is the installation of STATCOM type equipment [11–13].

In Ref. [14] a comparison is presented regarding the application of the solutions described above with the aim of choosing the right reactive power compensation solution depending on the particular situations in practice.

The LabVIEW environment [15] allows the monitoring and analysis of operating regimes, respectively, of the various phenomena that occur in the field of electrical networks [16, 17]. Thus, in Ref. [16] a solution for monitoring the electrical parameters of a single-phase system is presented, where the recorded data are written in files. In Ref. [17] a virtual tool is proposed for the analysis of the operating regimes of an electric power system. Both the block diagrams in which the virtual instruments are built, as well as the front panel with the results corresponding to the root mean square (RMS) values and the voltages and currents waveforms, respectively active, reactive, and apparent powers, are presented here.

Another advantage of using LabVIEW is the fact that it allows the creation of intuitive interfaces in process analysis [18, 19]. Also, LabVIEW can be integrated with MATLAB in order to model, simulate, and test processes [18, 20, 21].

The chapter proposes a LabVIEW virtual instrument that, by means of National Instruments DAQ devices, monitors the operating conditions of a three-phase load and commands a three-phase compensator for power factor improvement.

Within this chapter is presented a short theory necessary to understand the basic theoretical aspects regarding the power factor, respectively, the capacitive compensation of reactive power. Then, we proceed to the presentation of the automatic compensation algorithm and the implementation solution in LabVIEW (architecture of the virtual instrument). In order to test the virtual instrument, a laboratory model was built consisting of a RL consumer and a capacitive compensator in delta connection. The inductive component of the load is adjustable in steps on each phase and is commanded within the automated process. The capacitive compensator is also adjustable in steps, with the possibility of single-phase control. National Instruments equipment compatible with LabVIEW software was used for controlling the adjustable components of the laboratory model and analyzing the process. A stand-alone real-time application was build and deployed on the cRIO 9040 controller in order to realize an embedded operating system.

2. Power factor: Theoretical aspects

Considering a dipolar circuit (**Figure 1a**), the product between the instantaneous values of the voltage at the source terminals, u_s, and the current is called the instantaneous power (Eq. (1)) and can be visualized in **Figure 1b**.

$$p = u_s \bullet i \tag{1}$$

In relation to the active power, it is important to reveal the dependence of this quantity on the phase shift φ. At the same RMS values of voltage and current, the active power varies within wide limits with φ. For example, in **Figure 2a**, the case $\varphi = 0$ was considered, and in **Figure 2b** $\varphi = \pi/2$ was considered (extreme cases). In the first case, $\varphi = 0$ (resistive circuit), the active power is the highest ($P = U_s \cdot I$). It can also be noted that the instantaneous power has only positive values. If $\varphi = \pi/2$ (ideal coil—inductive circuit), the active power is zero, $P = 0$ and therefore, the instantaneous power oscillates between the circuit and the source. The same result is obtained in the case of an ideal capacitor ($\varphi = -\pi/2$).

The RMS value of the active power in a single-phase electric circuit is calculated in Eq. (2).

$$P = U_s \bullet I \bullet \cos\varphi \tag{2}$$

The factor $cos(\varphi)$ that intervenes in the expression of the active power (Eq. (2)) is called the power factor. In Eq. (3) the active current (active component of the

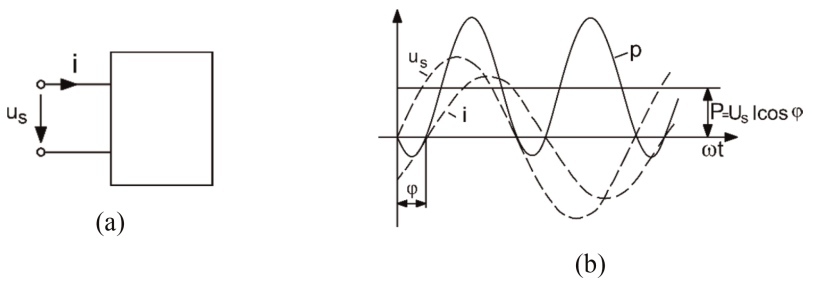


Figure 1.
Dipolar circuit (a) and instantaneous power in an electric circuit (b).

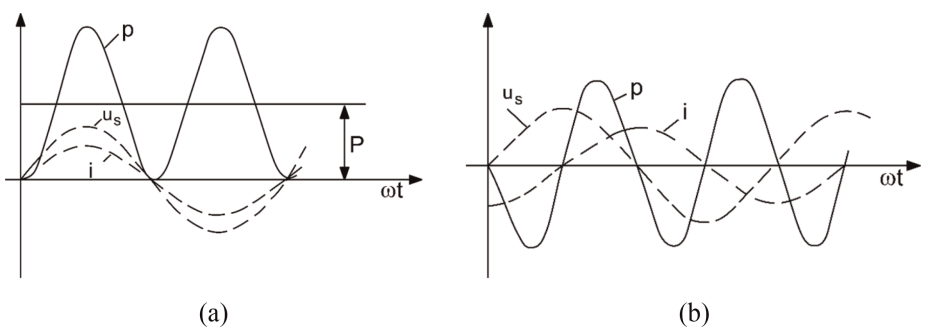


Figure 2.
Instantaneous power in a purely resistive circuit (a), respectively, in a purely reactive inductive circuit (b).

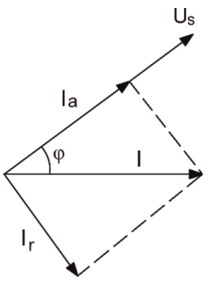


Figure 3.
The phasor diagram of the current.

current) is defined, which in a phasor diagram represents the projection of the current according to the direction of the voltage (**Figure 3**).

$$I_a = I \bullet \cos\varphi \tag{3}$$

The active power can therefore also be written in the form:

$$\mathrm{P} = U_s \bullet I_a \tag{4}$$

The component of the current in a direction perpendicular to the voltage at the terminals (**Figure 3**) is called the reactive component of the current and is defined using Eq. (5).

$$I_r = I \bullet \sin\varphi \tag{5}$$

Based on the reactive component of the current, the reactive power can be defined as follows:

$$Q = U_s \bullet I \bullet \sin\varphi = U_s \bullet I_r \tag{6}$$

In power systems, the power factor should be as high as possible. From Eq. (2), it follows that at a certain voltage at the terminals (the usual case of power supply networks), the same active power can be obtained at different currents. Of course, the solution to obtain a certain power (the required one) with as small current as possible, which means a high power factor, is the rational solution, as it involves small energy losses. A higher power factor can be obtained if the absolute value of the reactive power considered is small.

Unlike active power, reactive power can be positive or negative. Taking into account the rule of signs from receiver circuits, if the circuit is inductive, the reactive power is considered positive and is called inductive reactive power (Q_{ind}). In the situation where the circuit is capacitive, the reactive power is considered negative and is called capacitive reactive power (Q_{cap}).

Increasing the power factor can be achieved through a procedure called reactive power compensation. If an electric circuit absorbs reactive power, it can be compensated (reduced) by supplying reactive power of the opposite sign:

- an inductive reactive power can be reduced by adding a capacitive reactive power; therefore, an inductive electric circuit is compensated by a capacitor bank;

- a capacitive reactive power can be reduced by the contribution of an inductive reactive power, so that a capacitive electric circuit is compensated by coils.

The total reactive power (Q_t) remaining in the circuit after applying the compensation can be defined using Eq. (7)

$$Q_t = Q_{ind} - Q_{cap} \tag{7}$$

In three-phase circuits, the reactive power is calculated as the sum of the reactive powers on the three phases of the circuit:

$$Q = U_1 \bullet I_1 \bullet \sin\varphi_1 + U_2 \bullet I_2 \bullet \sin\varphi_2 + U_3 \bullet I_3 \bullet \sin\varphi_3 \tag{8}$$

The three-phase capacitor bank is composed of an assembly of single-phase units connected together to form a three-phase connection system. From the point of view of connecting single-phase units in a three-phase bank, it is possible to connect them in delta or star. The delta connection scheme has the advantage that, for the same number of capacitors, the reactive power supplied (Q_{CDelta}) is three times higher than in the case of star connection (Q_{CY}). The capacitive reactive power of the capacitor bank is determined according to their connection and thus results the value of the capacitance (C_Y or C_{Delta}) required to be installed in the capacitor bank to achieve total compensation:

$$C_Y = -\frac{Q_{CY}}{\omega \bullet U^2} \tag{9}$$

$$C_{Delta} = -\frac{Q_{CDelta}}{3 \bullet \omega \bullet U^2} \tag{10}$$

3. Hardware device presentation

The virtual instruments developed in LabVIEW allow the use of data acquisition equipment to create real-time applications both in the field of education and in the industrial field. This subchapter presents the hardware equipment used to test the developed virtual instruments.

3.1 NI cRIO-9040 controller

The cRIO-9040 is a rugged, high-performance, customizable embedded controller that offers a dual-core processing. The controller provides precise, synchronized timing and deterministic communications, which is ideal for highly distributed measurements. This controller offers several connectivity ports, which allow to add a local human machine interface and program, deploy, and debug software, which simplifies application development. The chassis of the cRIO-9040 can be used with a combination of C-series I/O modules to create a combination of analog and digital I/O measurements (**Figure 4**) [22].



Figure 4.
NI cRIO-9040 controller.



Figure 5.
NI 9242 module.

3.2 NI 9242 module

The NI-9242 module, shown in **Figure 5**, has four analog measurement channels for voltages up to 250 V rms, so it can be used for measurements in single-phase or three-phase systems. The wide measurement range makes it ideal for voltage measurement applications such as phase voltage measurement, power measurement, power quality monitoring. Transient and harmonic analysis can also be performed with simultaneous high-speed sampling [23].

3.3 NI 9227 module

The NI-9227, shown in **Figure 6**, was designed to provide high-precision measurements to meet the demands of data acquisition and control applications. It includes built-in anti-aliasing filters. With four channels of analog inputs for currents up to 5A, if used in conjunction with a C-series voltage input module, NI-9227, it can measure the power consumption and energy for various applications. Different power quality indicators can also be determined: noise, frequency, and harmonics [24].

3.4 NI 9478 module

The NI 9478, shown in **Figure 7**, is a digital output module for CompactDAQ and CompactRIO systems. It features 16 output channels, each channel is compatible with

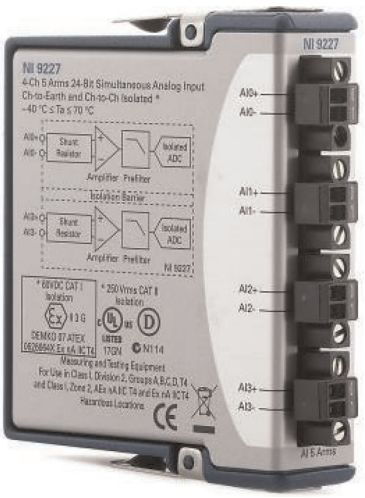


Figure 6.
NI 9227 module.

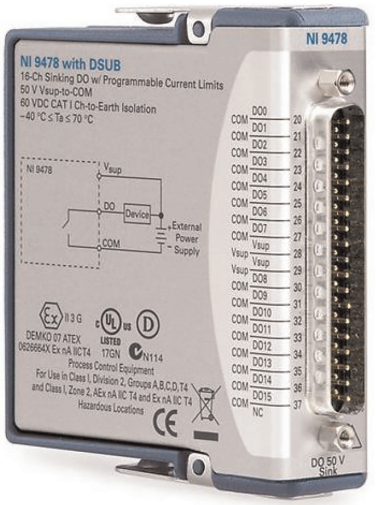


Figure 7.
NI 9478 module.

signals from 0 to 50 V. The NI 9478 works with industrial logic levels and signals to connect directly to a wide range of industrial relays [25].

The output pin numbering of the NI 9478 module is shown in **Figure 8**. Connection to the relays and power supply is made *via* the NI 9923 accessory. This terminal block is designed with a front connection with 37-pin input terminals (**Figure 8**). The NI-9923 facilitates optimal cable positioning and is supplied with a set of jack screws to ensure firm connectivity if intended to be used as part of a high vibration system. The pin terminals of the NI-9923 can be easily accessed after removing the four securing screws of the cover.

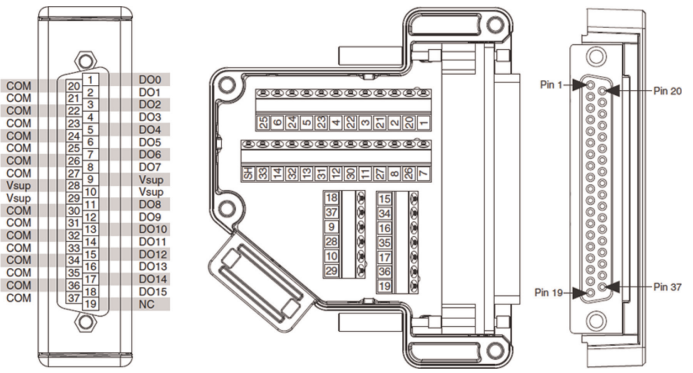


Figure 8.
Output pins of the NI 9478 module and the NI 9923 terminal block.

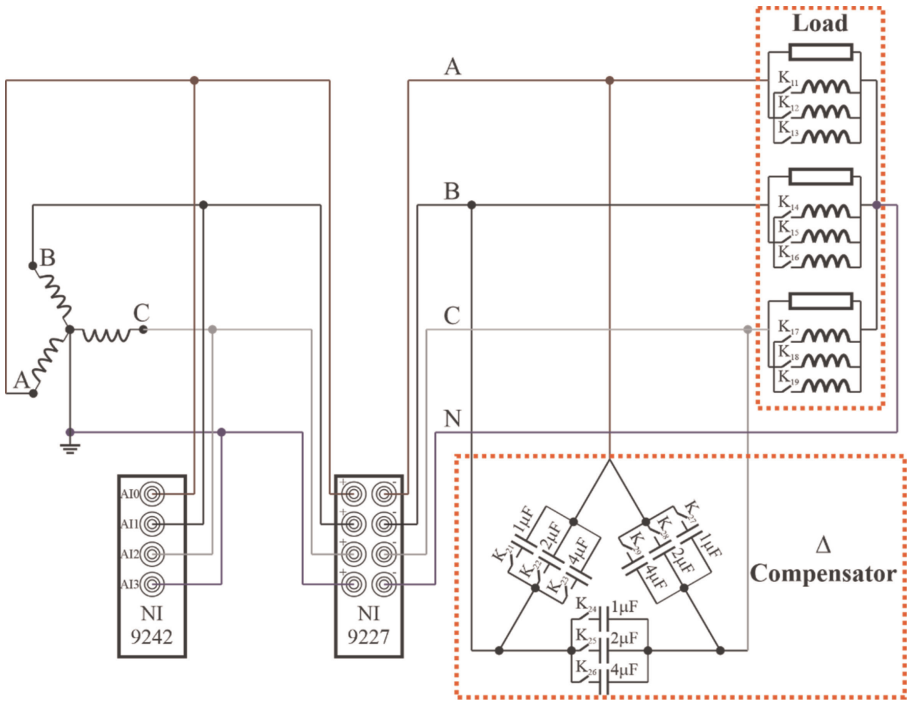


Figure 9.
Modeling diagram of the consumer and the reactive power compensator.

4. The laboratory model built for the compensator control

A relevant example of using LabVIEW in industry and education is the automatic control of a shunt reactive power capacitive compensator. The compensator proposed in this chapter is composed of three compensation steps, in Delta connection, with different values of the capacities: 1 μF, 2 μF, and respectively 4 μF. To test the virtual instrument for compensator control, a laboratory model consisting of an RL consumer in Y connection, powered at 400 V, phase to phase voltage, shown in **Figure 9**, was made.

For both inductive component of the consumer and capacitances of the compensator, the command relays are presented. The correctness of the operation of the compensation algorithm implemented in LabVIEW is verified by monitoring the consumer-compensator assembly, at the common point of connection to the network, using the current and voltage acquired by NI 9227 and NI 9242.

The possible combinations of symmetrical connection of the three steps lead to seven compensation regimes, according to **Table 1**. The table also shows the values of the compensation reactive power values and the equivalent capacity values corresponding to each regime.

The reactive inductive component of the load is step-adjustable, 293 mH, 488 mH, respectively 1465 mH, so as to obtain different values of the power factor and implicitly different values of the compensation requirement. The control of the different steps of the inductive consumer is done from the NI 9478 equipment, according to **Figure 10**.

After the acquisition of currents and voltages by means of the NI 9227 and NI 9242 devices, the power factor and the reactive power required for compensation are determined in LabVIEW. Depending on the result obtained, the NI 9478 device will control the step (or steps) corresponding to the capacitor bank (**Figure 11**).

Regime	The connection variant of the three steps	Q_k [VAR]	C [μF]
1	1	150,796	1
2	2	301,593	2
3	3	603,185	4
4	1 + 2	452,389	3
5	1 + 3	753,982	5
6	2 + 3	904,778	6
7	1 + 2 + 3	105,557	7

Table 1.
Possible operating regimes of the reactive power compensator.

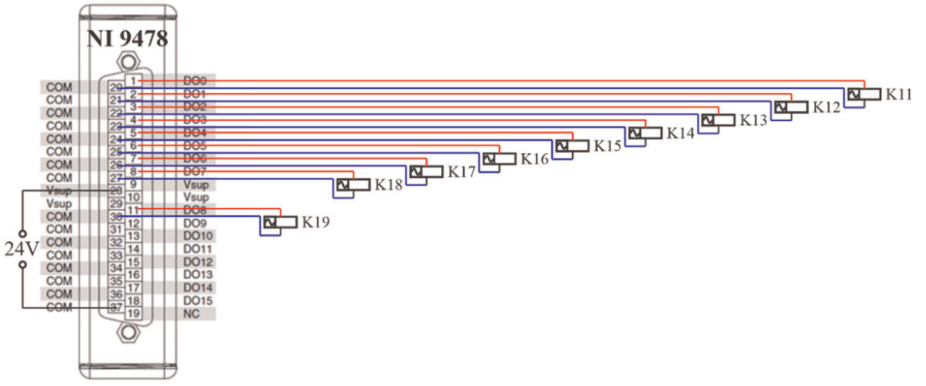


Figure 10.
The control-command diagram of the consumer.

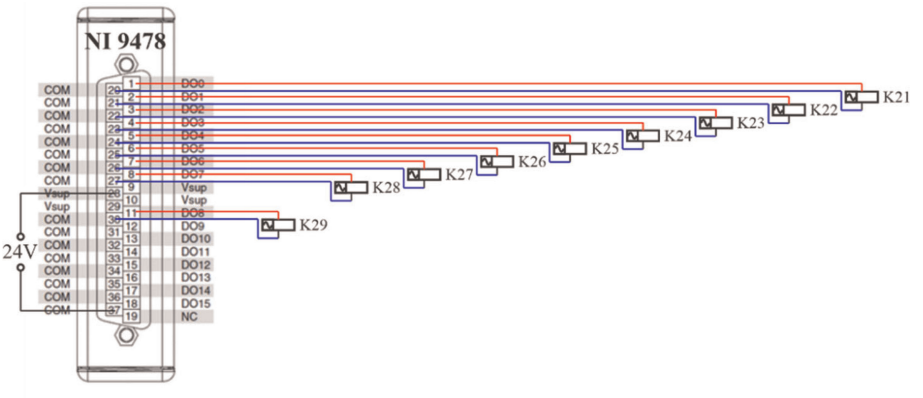


Figure 11.
The control-command diagram of the compensator.

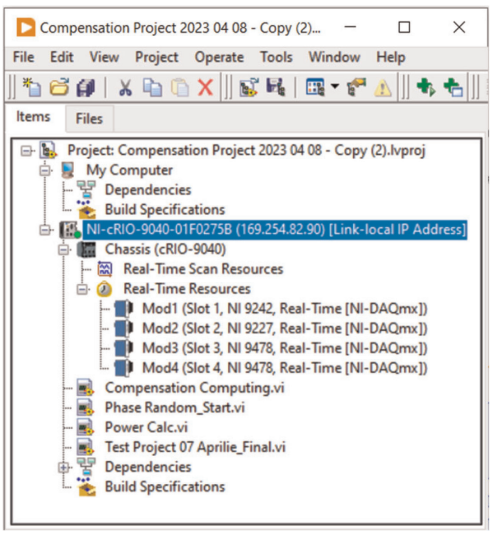


Figure 12.
The compensation project window.

5. Virtual instrument architecture

In order to implement the compensation algorithm, a LabView project was realized and a Real-Time application was build and deployed on the cRIO 9040. In **Figure 12**, the project window containing the real-time resources used is presented.

A *State Machine* design is used to implement the automated process in LabVIEW. The state diagram of the algorithm is presented in **Figure 13**. Here, it can be seen that the algorithm is carried out through five distinct states, which can be labeled *Step 0, Initialization, Process 1, Process 2,* and *Process 3*. In this subchapter is presented the block diagram of the virtual instruments developed with the aim of determining the compensation requirement and the control of the corresponding steps of the shunt capacitive compensator.

The LabVIEW *State Machine* used to apply the algorithm described in **Figure 13** consists of five Case Structures within a While loop, where each of the Case Structure's contains the code associated with one of the state machine's states.

Since it is an automatic process that should execute continuously depending on the load curve, the While loop executes continuously, and the Case structures repeat successively as shown below. The time delay imposed for each case structure is set to in order to be able to follow the development of the entire process and verify the correctness of the algorithm's operation.

a. The purpose of the "*Step 0*" case structure is to switch-off the capacitors each time before the entire process is restarted (**Figure 14**). This step is also useful due to the fact that the application also has a didactic character, and the operation of the algorithm and the development of the process can be easily followed by students.

b. The purpose of the "*Initialization*" case structure is to give a random command for the three steps of the coils corresponding to the three-phase consumer presented in **Figure 9**. *Phase Random_Start subVI* is called here for this purpose (**Figure 15**).

Its main element is the predefined function Random Number (0-1), which returns a random number between 0 and 1. By means of two simple mathematical tricks of

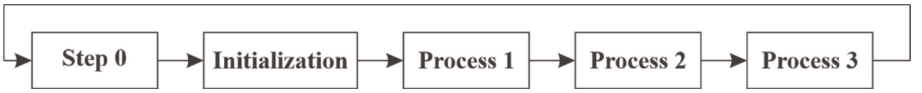


Figure 13.
The state diagram of the algorithm.

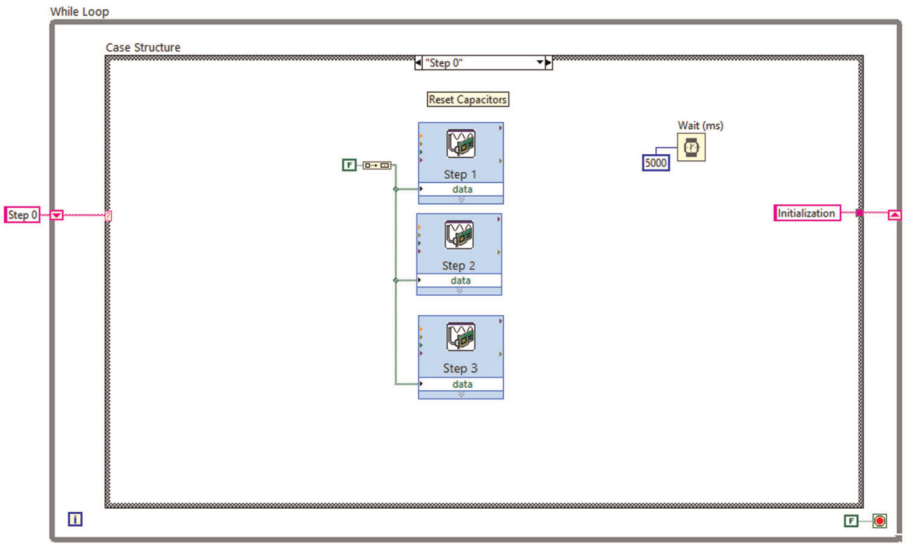


Figure 14.
"Step 0" *case structure Block diagram.*

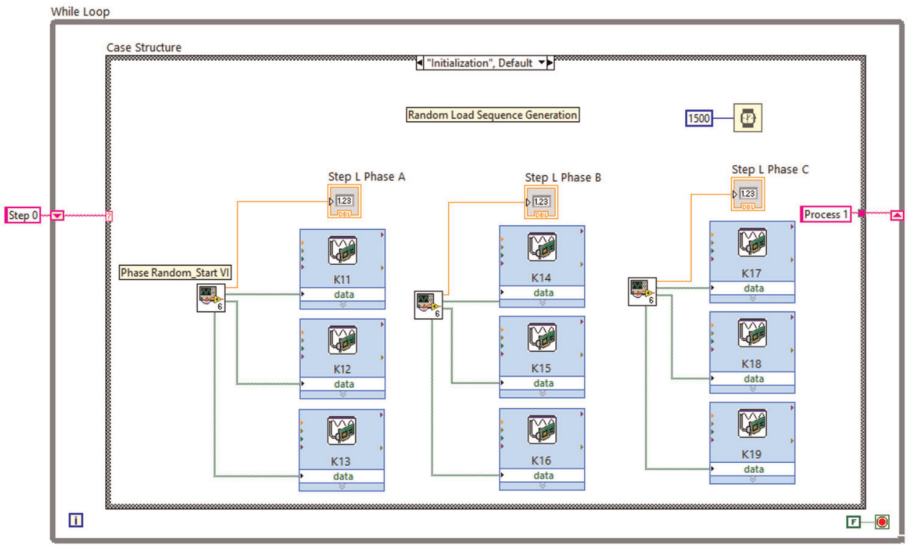


Figure 15.
"Initialization" *case structure Block diagram.*

multiplication and rounding, a random value between 1 and 3 corresponding to the three steps of the coil is then obtained. The procedure is repeated three times, corresponding to the three phases of the network, thus resulting in a balanced or unbalanced consumer depending on the random values provided at the input. When executing the "*Initialization*" structure, among the relays K11...K19, corresponding to the contacts of the three steps of the coils (three on each phase), **Figure 9**, only those corresponding to the generated random values, one for each phase, will receive a switching command *via* DAQ Assistant. The selection is done using the Boolean values resulted by comparison of the generated random values with the constants 1, 2, 3 through the Equal function from *Functions/Comparison*.

c. The "*Process 1*" case structure, **Figure 16**, calls the *Power Calc subVI* and fulfills, first of all, the function of acquiring the voltage and current waveforms on the three phases. It displays the active and reactive powers per phase. The reactive powers are the inductive reactive powers of the load phases. Within *Power Calc subVI,* by means of the predefined Extract Single Tone Information VI, the amplitude and phase of the fundamental frequency current and voltage signals are determined, and then, by dividing the amplitude by $\sqrt{2}$, their rms values. The voltage and current phase values are entered as input quantities in a FOR loop within which, using a Formula Node structure, a procedure for determining the phase shift between current and voltage is applied. This procedure is necessary because the acquired phase values have values in the range [−360, +360] degrees. Having the RMS values of voltage and current and the phase shift between them, the active power, the reactive power, and the power factor can be determined. The procedure is repeated three times, corresponding to the three phases of the network. Taking into account the fact that we propose to compensate the reactive power, also in this case structure the sum of the reactive powers on the three phases is made. The

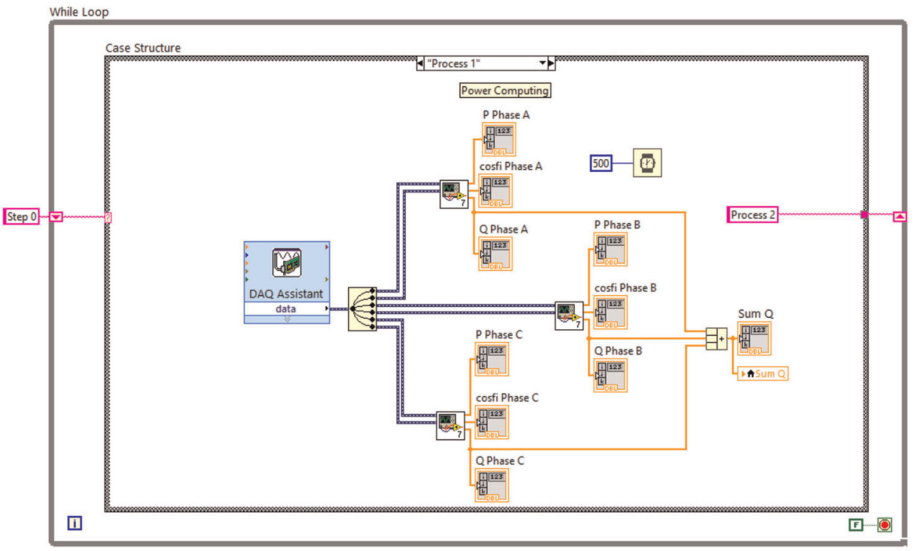


Figure 16.
"Process 1" *case structure Block diagram.*

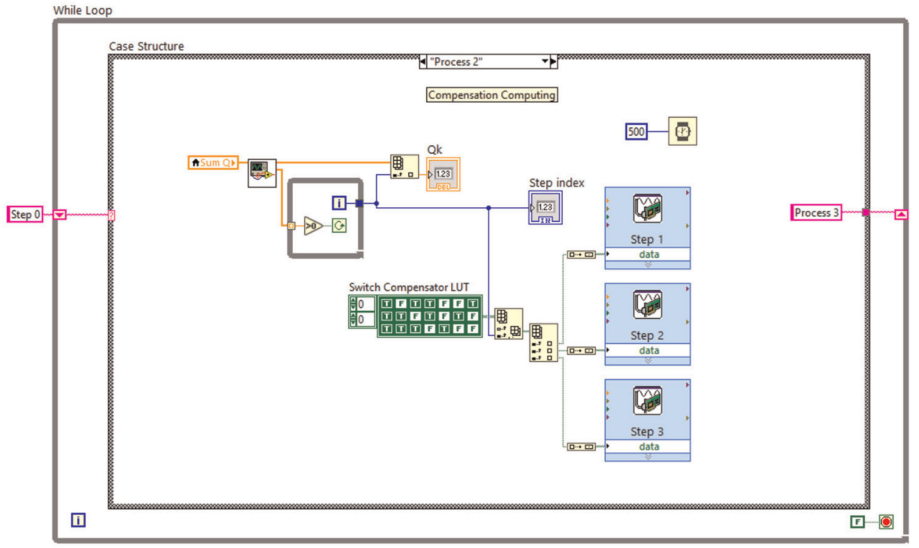


Figure 17.
"Process 2" *case structure Block diagram.*

obtained value will be transferred, by means of the *SumQ* local variable, to the next case structure.

d. Within the Case Structure "*Process 2*," **Figure 17**, the *Compensation Computing subVI* is called and the value of the compensation capacitive reactive power, *Qk*, is determined. The value of the reactive power required for the compensation is read from the local variable *SumQ*, obtained within "*Process 1*" case structure, is compared with the values of the capacitive reactive powers installed on each

step of the compensator. The step having the value closest to the required compensation power value, without going into capacitive overcompensation regime, will be connected. For the presented application, there are three steps of capacitors and the possible combinations result in seven steps of compensation. Choosing the right compensation step is done using the *Lookup Table* function in LabVIEW. In this case, the table contains Boolean values, which becomes input data for closing the compensator relays.

e. Within the case structure "Process 3," **Figure 18**, the reactive load is switched off in order to prepare to restart the automatic process.

6. Results

In order to test the correctness of the virtual instrument operating, three balanced and one unbalanced operating conditions are presented. The small differences between the active and reactive values measured on the load phases in the balanced operating regimes are due to the network voltage asymmetries that supply the laboratory model.

Figure 19 presents the powers and power factor values before and after the compensation corresponding to regime 1, the highest value of the reactive inductive power.

Figure 20 presents the voltages waveforms corresponding to the three phases and the phase shift between the voltage and the current corresponding to phase A before and after the compensation.

Being a highly inductive regime, we observe that the voltages waveforms are slightly distorted and also it can be seen that after the compensation, the current is more distorted than before the compensation. In this regime, the capacitive

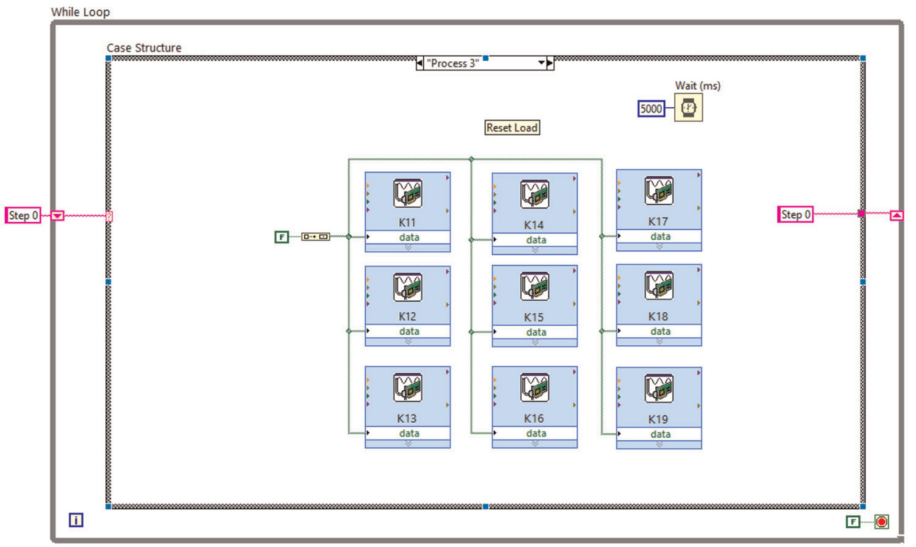


Figure 18.
"Process 3" *case structure Block diagram.*

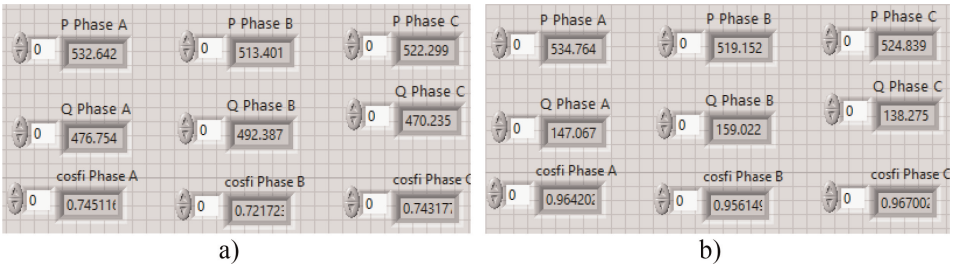


Figure 19.
Regime 1 operating conditions (highest inductive reactive load): (a) before the compensation; (b) after the compensation.

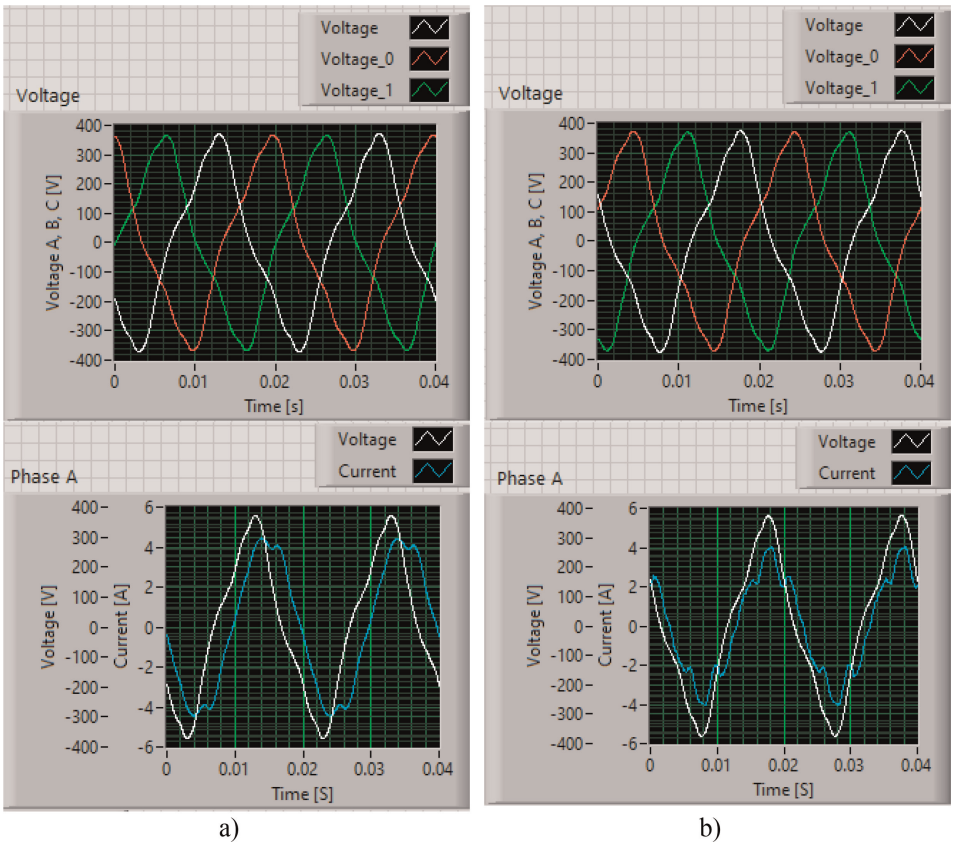


Figure 20.
Current and voltage waveforms—Regime 1: (a) before the compensation; (b) after the compensation.

compensator is operating on the highest step and the power factor after the compensation is about 0.96.

Figure 21 presents the powers and power factor values before and after the compensation corresponding to regime 2, the medium value of the reactive inductive power.

Figure 22 presents the voltages waveforms corresponding to the three phases and the phase shift between the voltage and the current corresponding to phase A before and after the compensation.

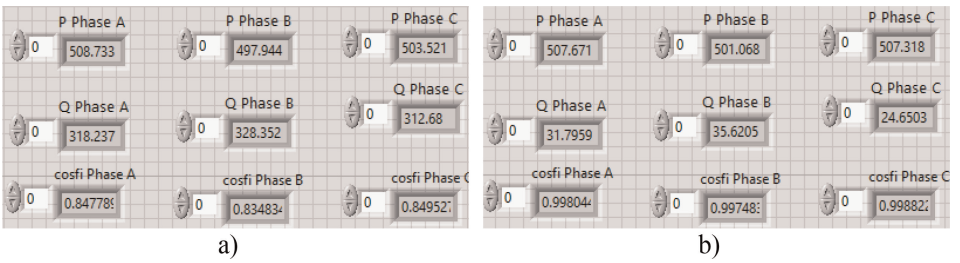


Figure 21.
Regime 2 operating conditions (medium inductive reactive load): (a) before the compensation; (b) after the compensation.

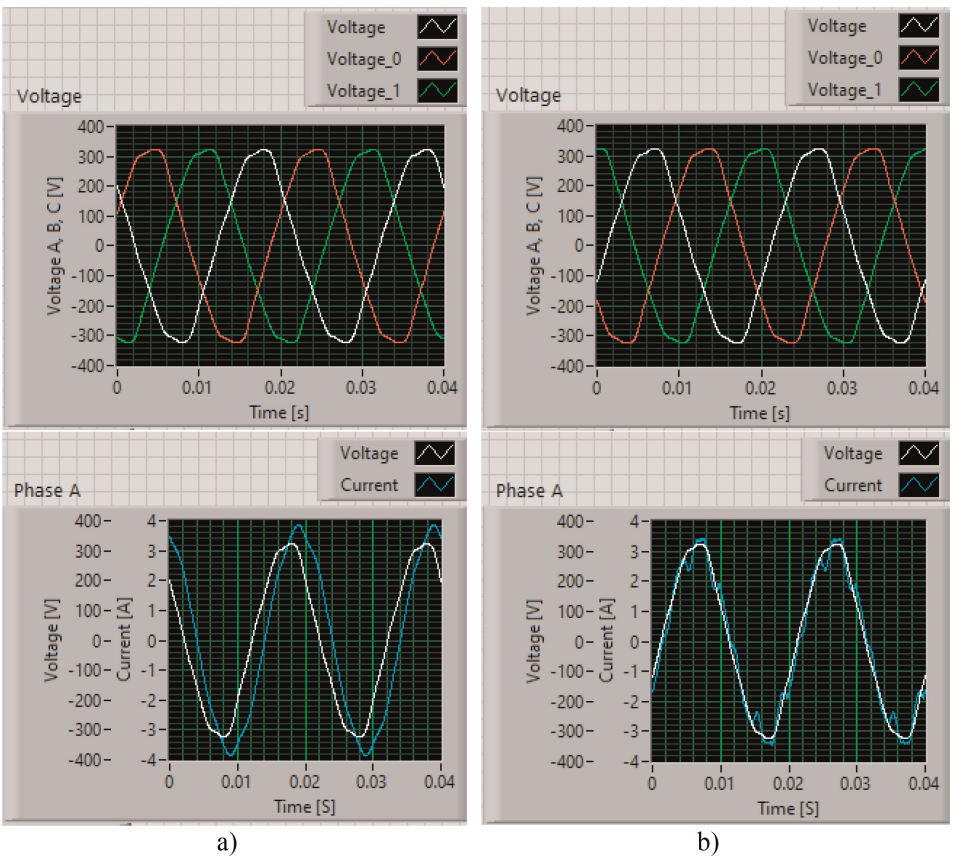


Figure 22.
Current and voltage waveforms—Regime 2: (a) before the compensation; (b) after the compensation.

This time, the voltages waveforms are no more so distorted like in the precedent case and the capacitive compensation is total, power factor is practically 1, but the current remains distorted after the compensation due to the presence of the capacitors.

Figure 23 presents the powers and power factor values before and after the compensation corresponding to regime 3, the lowest value of the reactive inductive power.

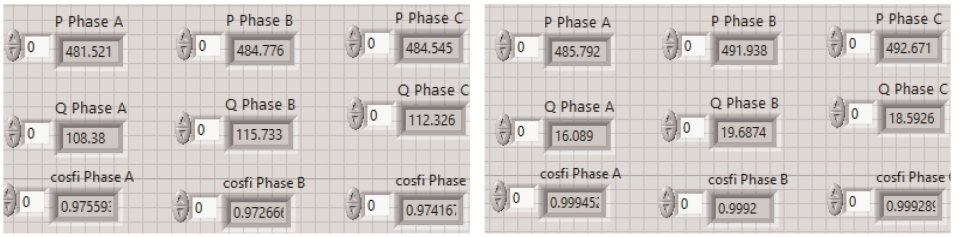


Figure 23.
Regime 3 operating conditions (lowest inductive reactive load): a) before the compensation; b) after the compensation.

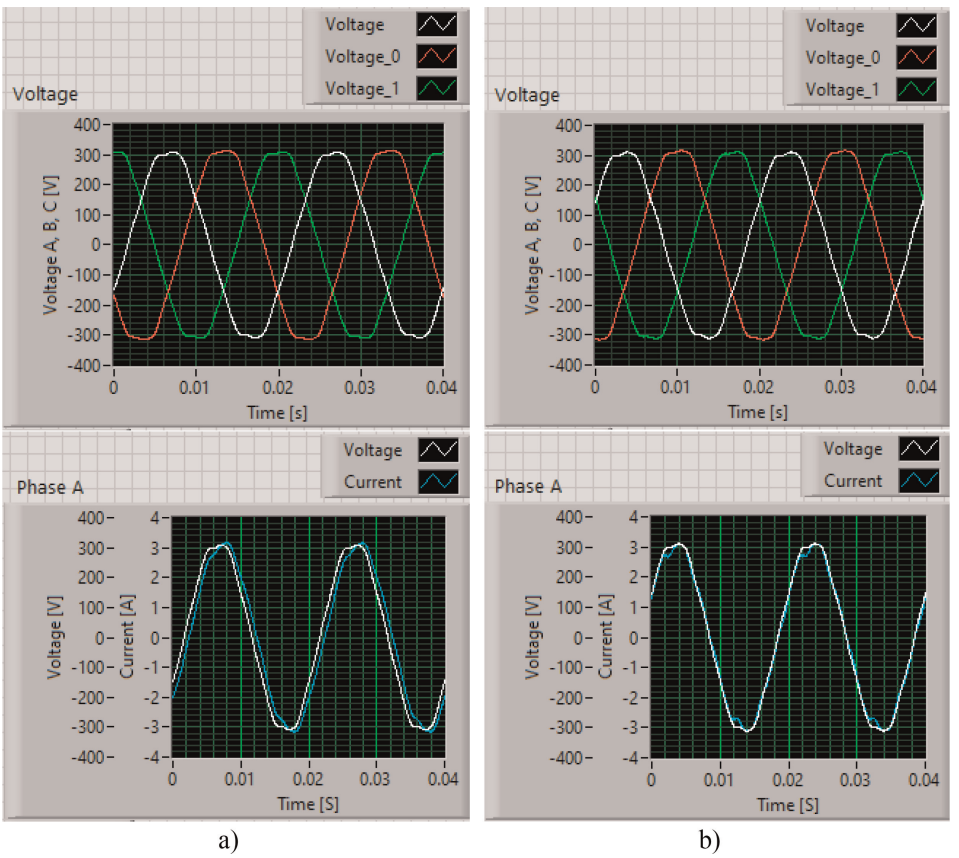


Figure 24.
Current and voltage waveforms—Regime 3: (a) before the compensation; (b) after the compensation.

Figure 24 presents the voltages waveforms corresponding to the three phases and the phase shift between the voltage and the current corresponding to phase A before and after the compensation.

In this case, the capacitive compensation is total, power factor is practically 1, and, due to the fact that the capacitive compensator intervenes with the lowest step, the current is almost sinusoidal, without significant distortions.

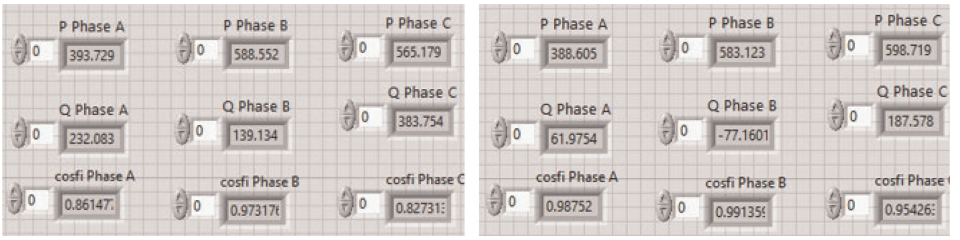


Figure 25.
Regime 4—Unbalanced operating conditions: (a) before the compensation; (b) after the compensation.

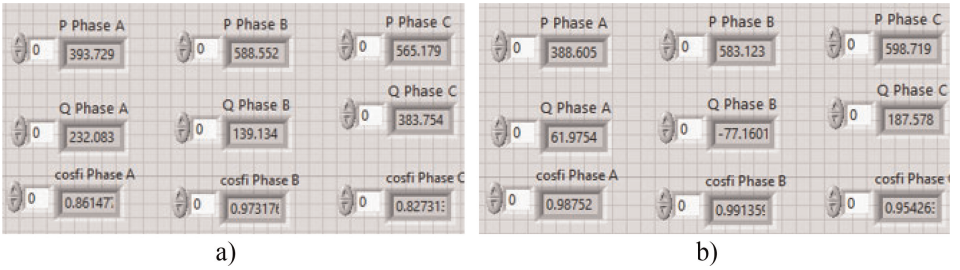


Figure 26.
Current and voltage waveforms—Unbalanced regime, before the compensation.

An unbalanced operating regime (Regime 4) is also presented. In this case, the inductive load operates on the medium step on the A phase, on the lowest step on the C phase and on the highest step on the B phase. The values of the active and reactive powers and the power factor are presented in **Figure 25**.

Figure 26 presents the voltages and the currents waveforms before the compensation. It can be observed in **Figure 26** that the inductive load unbalance causes voltage asymmetries and different power factor on the three phases, and this can be also remarked in the phase shift between voltage and current on each phase.

Figure 27 presents the voltages and the currents waveforms after the compensation. It can be observed that on the phase B is a capacitive overcompensation (negative reactive power), but overall, on the three phases, the total reactive power is inductive (positive), which was the goal of the virtual instrument presented here.

7. Conclusions

Capacitor banks are widespread in electrical distribution networks, this being the cheapest solution for power factor correction.

The accomplishment of a capacitive compensator for power factor correction is not a novelty in power systems, but using LabVIEW software and National Instrument technology offers important perspectives regarding the loads operating regime analysis and optimization. National Instruments technology, through cRIO system and digital output modules, allows obtaining a large number of outputs. This is particularly important when using step switching for large loads that have several steps of capacitors installed.

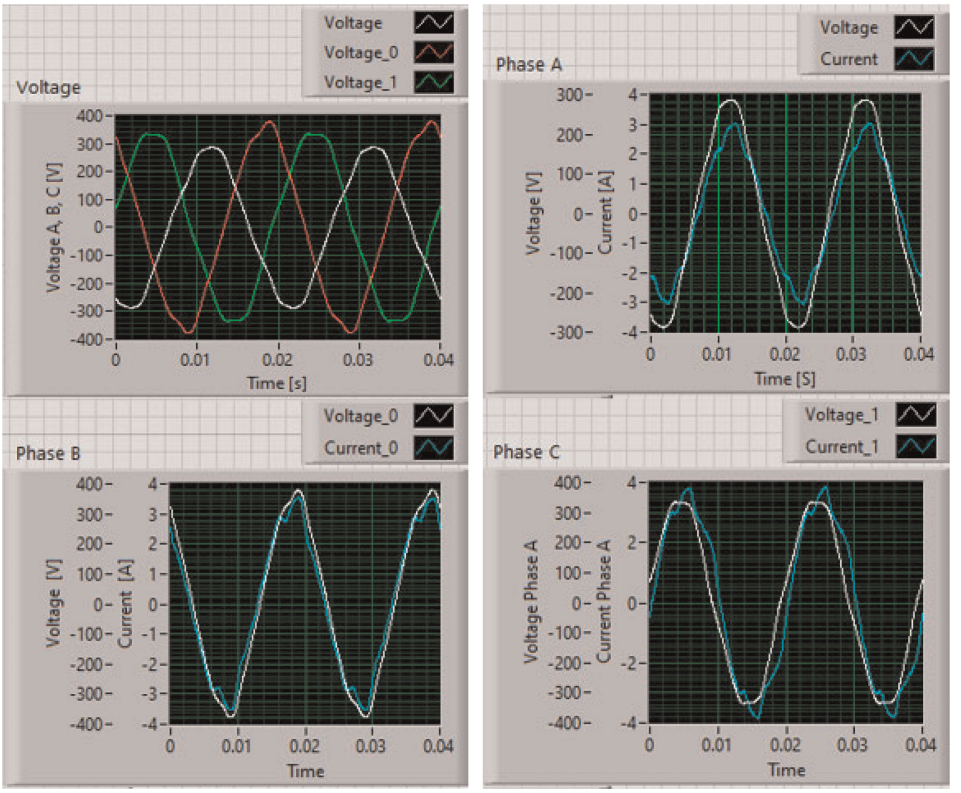


Figure 27.
Current and voltage waveforms—Unbalanced regime, after the compensation.

The solution proposed in this chapter can be implemented especially for unbalanced capacitive compensators designed for load balancing, where the control is done separately on each phase and a larger number of capacitor steps allow a finer control and therefore a better load balancing.

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References

[1] Mahdavi Tabatabaei N, Jafari Aghbolaghi A, Bizon N, Blaabjerg F. Reactive Power Control in AC Power Systems: Fundamentals and Current Issues. 1st ed. Germany: Springer, Power Systems; 2017. p. 634. DOI: 10.1007/978-3-08 319-51118-4

[2] Pana A, Baloi A, Molnar-Matei F. From the balancing reactive compensator to the balancing capacitive compensator. Energies. 2018;**11**(8):1979. DOI: 10.3390/en11081979

[3] Erxia L, Wanxing S, Xiaojun W, Bin W. Combined compensation strategies based on instantaneous reactive power theory for reactive power compensation and load balancing. In: 2011 International Conference on Electrical and Control Engineering. Yichang, China: IEEE; 16-18 September 2011. DOI: 10.1109/ICECENG.2011.6057765

[4] Lee SY, Wu CJ, Chang WN. A compact control algorithm for reactive power compensation and load balancing with static VAr compensator. Electric Power Systems Research. 2001;**58**:63-70. DOI: 10.1016/S0378-7796(01)00127-4

[5] Mayordomo JG, Izzeddine M, Asensi R. Load and voltage balancing in harmonic power flows by means of static VAr compensators. IEEE Transactions on Power Delivery. 2002;**17**:761-769. DOI: 10.1109/TPWRD.2002.1022801

[6] Grünbaum L, Petersson A, Thorvaldsson B. FACTS Improving the Performance of Electrical Grids. ABB Review (Special Report on Power Technologies). Zürich, Switzerland: ABB Group; 2003. pp. 13-18

[7] Dixon J, Morán L, Rodríguez J, Domke R. Reactive power compensation technologies, state of-the-art review. Proceedings of the IEEE. 2005;**93**: 2144-2164. DOI: 10.1109/JPROC.2005.859937

[8] Quintela FR, Arevalo JMG, Redondo RC. Power analysis of static VAr compensators. International Journal of Electrical Power & Energy Systems. 2008;**30**:376-382. DOI: 10.1016/j.ijepes.2007.12.00

[9] Said IK, Pirouti M. Neural network-based load balancing and reactive power control by static VAR compensator. International Journal of Computer and Electrical Engineering. 2009;**1**:25-31. DOI: 10.7763/IJCEE 2009.V1.5

[10] Xu Y, Tolbert LM, Kueck JD, Rizy DT. Voltage and current unbalance compensation using a static VAr compensator. IET Power Electronics. 2010;**3**:977-988. DOI: 10.1049/iet-pel.2008.0094

[11] Haque MH. Compensation of distribution system voltage sag by DVR and DSTATCOM. In: 2001 IEEE Porto Power Tech Proceedings (Cat. No.01EX502). Porto, Portugal: IEEE; 10-13 September 2001. DOI: 10.1109/PTC.2001.964609

[12] Capitanescu F, Wehenkel L. Redispatching active and reactive powers using a limited number of control actions. IEEE Transactions on Power Systems. 2011;**26**:1221-1230. DOI: 10.1109/TPWRS.2010.2102371

[13] Roncero-Sànchez P, Acha E. Design of a control scheme for distribution static synchronous compensators with power-quality improvement capability. Energies. 2014;**7**(4):2476-2497. DOI: 10.3390/en7042476

[14] Xue Q, Jiang B, Chunxia C, Hang L. Comparison and analysis of reactive

power compensation strategy in power system. In: 2019 IEEE Sustainable Power and Energy Conference (iSPEC). Beijing, China: IEEE; 21-23 November 2019. DOI: 10.1109/iSPEC48194.2019.8975301

[15] Essick J. Hands-on Introduction to LabVIEW for Scientists and Engineers. 4th ed. New York: Oxford University Press; 2019. p. 702

[16] Neeraj K, Sonal J. Development of LabVIEW based electrical parameter monitoring system for single phase supply. In: 2015 Communication, Control and Intelligent Systems (CCIS). Mathura, India: IEEE; 07-08 November 2015. DOI: 10.1109/CCIntelS.2015.7437964

[17] Mohammad A, Ahmad AJ, Tariq EH. Power system analysis using LabVIEW. Universal Journal of Electrical and Electronic Engineering. 2020;7(6): 299-306. DOI: 10.13189/ujeee.2020.070601

[18] Iresha P, Sertac B. Development of LabVIEW based monitoring system for AC microgrid systems. In: 2018 IEEE 12th International Conference on Compatibility, Power Electronics and Power Engineering (CPEPOWERENG 2018). Doha, Qatar: IEEE; 10-12 April 2018. DOI: 10.1109/CPE.2018.8372604

[19] Prabhavathy M, Ramesh B, Kalpalatha Reddy T. An alternative distributed control using LabVIEW. In: 2014 International Conference on Computation of Power, Energy, Information and Communication (ICCPEIC). Chennai, India: IEEE; 16-17 April 2014. DOI: 10.1109/ICCPEIC.2014.6915361

[20] Zhang Z, Jiang H, Zhang H, Wang J. Active power filter design and simulation by combining LabVIEW and Simulink. In: 2011 International Conference on Advanced Power System Automation and Protection. Beijing, China: IEEE; 16-20 October 2011. DOI: 10.1109/APAP.2011.6180971

[21] Lokesh Reddy M, Indragandhi V, Kushal B, Rajasingh R. Integration of MATLAB and LabVIEW for motor control test bench with power analysis. In: 2019 Innovations in Power and Advanced Computing Technologies (i-PACT). Vellore, India: IEEE; 22-23 March 2019. DOI: 10.1109/i-PACT44901.2019.8960064

[22] NI cRIO-9040 Specifications. 2022. Available from: https://www.ni.com/docs/en-US/bundle/crio-9040-specs/page/overview.html [Accessed: June 5, 2023]

[23] NI 9242 Specifications. 2022. Available from: https://www.ni.com/docs/en-US/bundle/ni-9242-specs/resource/376130b_02.pdf [Accessed: September 12, 2022]

[24] NI-9227 Operating Instructions and Specifications. 2022. Available from: https://www.ni.com/docs/en-US/bundle/ni-9227-seri/resource/375101e.pdf [Accessed: September 12, 2022]

[25] NI-9478 Specifications. 2022. Available from: https://www.ni.com/docs/en-US/bundle/ni-9478-specs/page/specifications.html [Accessed: September 12, 2022]